

Spatially blocked pluvial flood learning on the H3 grid: open labels, adaptive refinement, and a rainfall-conditioned cell index

Working manuscript (methods orientation, IJDRR-shaped).

Abstract

Intense short-duration rainfall can overwhelm urban drainage and produce pluvial flooding with limited warning, and city-scale screening increasingly relies on machine learning and multi-resolution grids. Many operational indices, however, are tied to proprietary insurance labels and to evaluation protocols that ignore spatial leakage. This study presents an open-label machine-learning protocol for hexagonal pluvial flood screening on the H3 discrete global grid. Multi-source public flood indicators are joined to H3 cells, gradient-boosting classification and continuous-risk regression models are fitted, and performance is assessed with H3-block spatial cross-validation that withholds entire parent cells. The protocol also quantifies scale loss through a Jaccard/F1 hotspot ladder and screens parents for adaptive refinement using trained cell scores. The rainfall-conditioned hexagon index $PFI_h(c, r)$ is defined as a model output—distinct from feature importance and from the proprietary building-level PFI_b product. On a smaller Manhattan pilot ($n = 141$), spatial cross-validation accuracy is 0.784 ± 0.069 with F1 0.866, but the positive class dominates (80% prevalence) and a trivial always-positive classifier reaches 0.808 accuracy and 0.893 F1; out-of-fold ROC-AUC is 0.68 and average precision 0.86. On a larger pilot ($n = 956$, 47.9% positive), accuracy is 0.642 ± 0.148 , exceeding both the always-positive (0.479) and the constant majority-class (0.521) baselines, with continuous-risk R^2 of 0.525 ± 0.112 ; out-of-fold ROC-AUC is 0.70 and average precision 0.72. Adaptive refinement uses approximately 57% as many cells as a uniform fine grid. The results demonstrate the protocol under the stated boundaries; they do not establish citywide skill or rainfall-conditioned discrimination, because event rainfall remains a synthetic constant and the scenario response is flat.

Keywords: pluvial flood; H3; discrete global grid; spatial cross-validation; machine learning; flood susceptibility.

1 Introduction

Extreme rainfall events are increasing in frequency and intensity, and pluvial flooding—the flooding that occurs when intense rain overwhelms urban drainage before it can enter watercourses—can develop rapidly and with limited warning (Rosenzweig et al., 2021). Assessing this risk at city scale requires representations

that are computationally scalable, updateable as new observations arrive, and evaluable without silent spatial leakage.

Discrete Global Grid Systems (DGGS), and the hexagonal H3 system in particular, provide nested cells that support multi-resolution aggregation and neighbourhood queries. Svelling et al. (2026) showed that a machine-learning building-level pluvial susceptibility index can be aggregated into H3 cells, reducing spatial query cost by roughly two orders of magnitude, and that hotspot sets can diverge sharply across resolutions (Jaccard similarity of about 0.14 between street-level and neighbourhood hotspots on their proprietary stack). That work establishes H3 as a communication and screening fabric, but it focuses on aggregating and communicating a proprietary index rather than on an open, spatially honest learning protocol for jurisdictions without access to insurance claims.

This study addresses the following question: can an open, multi-source label stack on H3 support spatially blocked evaluation, diagnose scale loss, and drive adaptive refinement with an explicit rainfall-conditioned cell index, without claiming reproduction of any proprietary product? Unlike Svelling et al. (2026), who aggregate a pre-existing building-level, insurance-derived index into H3 cells for scalable communication, this study learns directly from open flood observations on the H3 support, evaluates transfer across held-out H3 spatial blocks, and adapts the grid itself rather than merely re-aggregating a fixed index. The contribution is a methods protocol rather than a citywide operational map, and consists of three parts. First, an open-label H3 feature and label assembly procedure with explicit provenance for assembly mode, feature source, label source, and rainfall source. Second, primary evaluation through spatial H3-block cross-validation, with random splits relegated to diagnostics and class-prevalence baselines disclosed alongside accuracy and F1. Third, a scale-loss Jaccard/F1 ladder and an adaptive H3 refinement stage screened by trained cell scores, together with a binding definition of $PFI_h(c, r)$ that is neither feature importance nor PFI_b .

Throughout, $PFI_h(c, r)$ denotes a rainfall-conditioned hexagon flood probability produced by the trained model on the H3 support. The symbol is deliberately reused from Svelling et al. (2026), where PFI_h instead denotes an H3-aggregated building-level index; the two quantities differ in training data (open flood observations versus insurance-derived building scores), in construction (direct H3-native learning versus post-hoc aggregation), and in meaning, and are therefore not equivalent. Claims of citywide skill, radar-based rainfall, parity with PFI_b , or rainfall-conditioned discrimination are outside the scope of the present evidence.

2 Related work

From building indices to hexagonal screening. Svelling et al. (2026, IJDRR) aggregate a machine-learning building-level pluvial susceptibility index (PFI_b) into H3 cells, reducing geometry-heavy spatial query cost by roughly two orders of magnitude while exposing a resolution trade-off: fine street-level hotspots are largely invisible at coarser neighbourhood grids (Jaccard about 0.14 between their fine and intermediate hotspot sets). A related preprint develops the same H3 indexing narrative (Svelling et al., SSRN). This line of work treats H3 as a scalable communication fabric for proprietary indices; its focus is aggregation and communication rather than open-label learning or spatially blocked evaluation. It is therefore the closest

precedent and an explicit boundary for the present study, which does not reproduce PFIb and does not equate its own Jaccard values to theirs.

Hexagonal DGGS as an analysis substrate. Independently of insurance labels, Li et al. (2022) use an ISEA3H hexagonal DGGS for multi-scale flood mapping under climate scenarios, emphasising resolution-consistent predictors. Their contribution supports DGGS nesting as a scientific substrate. The present study differs in its emphasis on an H3 learning and evaluation protocol for open urban flood indicators rather than on climate-scenario inundation mapping.

Spatial holdouts in GeoAI. Random train/test splits routinely inflate skill under spatial autocorrelation. Sun et al. (2023) establish the rationale for spatially separated validation in geospatial machine learning. This study operationalises that rationale with GroupKFold over coarse H3 parent identifiers, so that entire parent blocks are withheld and primary reporting aligns with leakage-aware practice rather than treating random accuracy as the headline metric.

Open urban flood observations and their biases. New York City 311 flooding complaints have previously supported statistical and machine-learning analyses of street flooding, and their reporting biases—who reports, and which flooded roads are driven—are explicitly documented (Agonafir et al., 2022a; Agonafir et al., 2022b). The present contribution is therefore not the use of open flood reports itself, but their assembly onto H3-native cells together with blocked spatial evaluation. **Urban pluvial machine-learning susceptibility.** City-scale studies map susceptibility from digital elevation models, land cover, and drainage proxies with classical machine learning (e.g., Bersabe & Jun, 2025, Seoul). Adaptive and non-uniform spatial resolution has separate antecedents in process-based flood simulation; here the grid is refined selectively from trained scores rather than prescribed by an inundation solver. The present study combines these strands—open multi-source labels, H3 nesting, adaptive refinement screened by trained scores, and an explicit rainfall-conditioned cell index—in a single reproducible pipeline with blocked cross-validation as the primary evaluation.

Relative to the PFIb-to-H3 aggregation papers, this work does not improve or reproduce a proprietary building index; it asks whether public flood indicators can support spatially blocked learning on H3. Relative to hexagonal DGGS flood mapping, it emphasises evaluation protocol and adaptive screening rather than scenario inundation alone. Relative to urban pluvial susceptibility maps, it adds H3-block cross-validation as the primary evaluation, an open-label scale-loss ladder, adaptive cell-count diagnostics, and a binding non-PFIb definition of $PFI_h(c, r)$, while withholding citywide, radar, and rainfall-discrimination claims until stronger evidence exists.

3 Study area and data

Extent. Two Manhattan pilot extents are used. The smaller is a Lower Manhattan bounding box (approximately 74.02–73.97°W, 40.70–40.76°N); the larger (`manhattan_expanded`) extends northward to approximately 74.03–73.94°W, 40.68–40.80°N. Both are pilot extents within New York City, not a citywide extent.

Layers. The following open layers were downloaded in August 2026: a digital elevation model (USGS 3DEP subset); DEP stormwater flood polygons; building footprints; USGS Hurricane Ida high-water marks; NYC 311 flooding points (ArcGIS/CDN mirrors); FEMA Sandy storm-surge inundation (negative control only); NLCD impervious fraction; and NHDPlus HR hydrography, from which `dist_stream_m` is derived as a distance-to-water proxy in a tidal and shoreline setting. FloodNet sensor data are available only as a stub and are not used. Event rainfall is a synthetic constant Ida-like hook rather than radar or gauge data.

Data sources. Elevation is from the USGS 3D Elevation Program; impervious fraction from the National Land Cover Database; hydrography from NHDPlus High Resolution; flood labels from the New York City Department of Environmental Protection stormwater flood polygons, NYC 311 flooding service requests, and USGS Hurricane Ida high-water marks; and the negative control from FEMA Sandy storm-surge inundation. Version and download dates are recorded in the repository download manifest. The assembly is open-data; rainfall may carry an event-raster provenance tag rather than a gauge or radar observation.

4 Methods

4.1 H3 representation

Training uses H3 resolution 9, configurable. Scale-loss diagnostics assemble labels at fine resolution 10 and roll hotspots to parent resolutions 9 and 8. Adaptive refinement expands selected parents to resolution 11. All joins use EPSG:4326.

4.2 Features and open labels

Static predictors are elevation, slope, a flow-accumulation proxy, impervious fraction, an urban land-cover flag, building density, and distance-to-water. Rainfall enters as rainfall intensity for scenario conditioning; observed static columns are tracked separately from rainfall so that a constant synthetic grid is not presented as radar.

Labels combine DEP polygon area fractions with 311 and Ida point counts into flood-risk and flood-class targets. These are observed flood labels, not ground-truth inundation, and PFIB is not used. FloodNet joins only when explicitly enabled and a non-empty layer exists.

4.3 Models and baselines

The primary learner is a gradient-boosting classifier with a continuous-risk regressor. Baselines are an L2-regularised logistic (or linear) model and an elevation–impervious–slope ponding rule. In-sample metrics are treated as optimistic references only. In addition, two constant classifiers—always-positive and always-negative—are computed on each held-out fold so that accuracy and F1 are never reported without a class-prevalence comparison; the constant majority-class classifier is derived from the pooled class count.

4.4 Spatial H3-block cross-validation

Cells are grouped by coarse H3 parents, and GroupKFold withholds entire blocks. Per-fold metrics are archived with each pilot run. For each held-out cell, the predicted class probability is retained, so that threshold-independent discrimination metrics (ROC-AUC and average precision, or PR-AUC) can be computed from out-of-fold predictions. Random independent splits are computed but are not primary.

4.5 Scale-loss diagnostics

Fine-resolution hotspot sets (the top quantile of risk) are compared with parent-aggregated hotspots using Jaccard similarity and F1 under mean, maximum, and p90 rollups. These values are not equated to Svelling et al.'s PFIb Jaccard of 0.14.

4.6 Adaptive refinement

After training, cell scores screen parents for refinement. The metrics are mixed-resolution cell counts relative to a fixed coarse grid and a uniform fine grid, expressed as an adaptive-to-uniform cell-count ratio. The pilot uses the trained PFI_h as the screening score.

4.7 Rainfall-conditioned index

The cell index is defined as

$$\mathrm{PFI}_h(c,r)=\widehat{P}(Y_c=1\mid X_c,r),$$

where static features (X_c) are held fixed while the rainfall condition (r) varies across named scenarios. This is a model output, not a SHAP or permutation importance, and not PFIb. Because the training rainfall is currently a constant, the model has not observed rainfall variation and $\mathrm{PFI}_h(c,r)$ cannot yet respond to it; the definition remains binding for future runs with observed rainfall.

4.8 Negative control

FEMA Sandy coastal inundation overlaps are reported for separation checks only and are never used as training labels.

5 Experimental design

ID	Description	Role
E1	Assemble the open Lower Manhattan table	Feature/label table for the pilot bbox
E2	Spatial cross-validation	Primary blocked accuracy / F1 and out-of-fold discrimination

E3	Baselines	Diagnostic comparison against constant and parametric classifiers
E4	Jaccard ladder	Scale-loss hotspot similarity
E5	Adaptive refinement and ablation	Cell-count comparison against uniform fine
E6	Rainfall scenarios	Within-cell PFI_h response across intensities
E7	Sandy negative control	Coastal overlap checks (not training)

Reproducible tables and figures are released with the public repository; process detail is documented in the accompanying research report.

6 Results

All figures are computed from the two Manhattan pilots and do not describe citywide performance.

6.1 Spatial H3-block cross-validation

Metric	Value
n_cells	141
spatial_cv_n_folds	5
spatial_cv_n_blocks	7
spatial_cv_accuracy_mean $\pm$ std	0.784 $\pm$ 0.069
spatial_cv_f1_mean	0.866
spatial_cv_r2_mean $\pm$ std	0.030 $\pm$ 0.343
spatial_cv_mae_mean	0.332
random_split_val_accuracy (diagnostic only)	0.690
positive-class prevalence (held-out)	0.801
always-positive (majority) baseline accuracy	0.808
always-positive (majority) baseline F1	0.893
always-negative baseline accuracy	0.192
spatial_cv_roc_auc_pooled	0.683
spatial_cv_pr_auc_pooled	0.861

Per-fold accuracy/F1: Fold0 0.755 / 0.850; Fold1 0.760 / 0.850; Fold2 0.773 / 0.872; Fold3 0.714 / 0.813; Fold4 0.917 / 0.944.

The held-out labels are highly imbalanced (80.1% positive). A trivial always-positive classifier would reach mean accuracy 0.808 and mean F1 0.893 across the same folds, which exceeds the model's 0.784 accuracy and

0.866 F1. Spatial blocking makes the evaluation design more defensible but does not by itself establish discrimination. The threshold-independent out-of-fold metrics are moderate: pooled ROC-AUC is 0.683 and pooled average precision is 0.861, the latter only slightly above the 0.801 prevalence baseline for a random ranker. Fold4 accuracy (0.917) coincides with a small test set ($n = 24$, two blocks) and is not interpreted in isolation. Continuous-risk R^2 (0.030 ± 0.343) is weak and reported for completeness. Random-split accuracy (0.690) remains diagnostic only.

6.2 Scale-loss Jaccard ladder (open labels)

Fine resolution R10, hotspot quantile 0.9:

coarse_res	aggregation	jaccard	f1
8	mean	0.167	0.286
8	max	1.000	1.000
8	p90	1.000	1.000
9	mean	0.977	0.988
9	max	1.000	1.000
9	p90	0.977	0.988

Under mean aggregation, parent rollup at R8 yields Jaccard 0.167 with fine R10 hotspot parents, consistent with strong scale smoothing on this open-label stack. Maximum and p90 aggregations retain extrema by construction and therefore should not be read as "no scale loss". These diagnostics do not reproduce Svelling et al.'s PFIb Jaccard of 0.14, which uses different labels, resolutions, and hotspot definitions.

6.3 Adaptive versus fixed and uniform fine grids

Field	Value
score used for screening	PFI_h
n_fixed_coarse (R9)	141
n_adaptive_mixed	3933
n_uniform_fine (R11)	6909
adaptive_cell_count_ratio (adaptive/uniform)	0.569
n_parents_refined	79
score_quantile	0.8

Adaptive mixed grids use about 57% as many cells as a uniform R11 grid (3933 versus 6909) while refining 79 of 141 coarse parents. Relative to the fixed R9 baseline the mixed grid uses about 27.9 times more cells

(3933 versus 141). This statement concerns cell counts only; wall-clock runtime, memory, and city-scale cost are not reported.

6.4 Rainfall scenarios

The scenario loop covers 141 cells across {moderate 25, heavy 40, ida_like 75, extreme 100 mm/h}; rainfall provenance remains an event-raster synthetic constant hook. The mean PFI_h is about 0.803 for every scenario, and the within-cell range across scenarios is 0. The current pilot therefore does not demonstrate rainfall-conditioned discrimination. The loop is correct (only rainfall_mm_h varies); the flat response follows from constant training rainfall, because all training cells carry rainfall_mm_h = 75 from the synthetic hook, so rainfall has zero training variance and a feature importance of 0. A non-zero response requires ingested observed event rainfall with multi-intensity, non-synthetic provenance and retraining.

6.5 Sandy negative control

Open-data assembly: n_cells = 141; n_coastal = 31; n_pluvial = 71; n_both = 23; n_coastal_only = 8; frac_coastal_only $\approx$ 0.057; pluvial-minus-coastal mean score $\approx$ 0.120. Coastal overlap is not a training label.

6.6 Expanded open-data pilot

Metric	Value
n_cells	956
spatial_cv_n_folds	5
spatial_cv_n_blocks	28
spatial_cv_accuracy_mean $\pm$ std	0.642 $\pm$ 0.148
spatial_cv_f1_mean	0.608
spatial_cv_r2_mean $\pm$ std	0.525 $\pm$ 0.112
spatial_cv_mae_mean	0.112
random_split_val_accuracy (diagnostic only)	0.667
positive-class prevalence (held-out)	0.479
always-positive baseline accuracy	0.479
always-positive baseline F1 (fold-mean)	0.641
constant majority-class (always-negative) baseline accuracy	0.521
constant majority-class (always-negative) baseline F1	0.000
spatial_cv_roc_auc_pooled	0.703
spatial_cv_pr_auc_pooled	0.723

Per-fold accuracy/F1: Fold0 0.801 / 0.832; Fold1 0.419 / 0.442; Fold2 0.759 / 0.736; Fold3 0.516 / 0.343; Fold4 0.715 / 0.689.

The expanded extent is a second, larger open-data pilot (956 cells over 28 blocks), still not citywide. Its held-out labels are near-even (47.9% positive), in contrast to the 80% prevalence of the smaller table. Under the same H3-block protocol, spatial cross-validation accuracy is 0.642 ± 0.148 , exceeding both the always-positive baseline (0.479) and the constant majority-class baseline (0.521). Out-of-fold discrimination is moderate: pooled ROC-AUC is 0.703 and pooled average precision is 0.723, the latter clearly above the 0.479 prevalence baseline. Mean F1 (0.608) nevertheless remains below the fold-mean always-positive F1 (0.641). The per-fold spread (accuracy 0.419–0.801; F1 0.343–0.832) coincides with heterogeneous held-out class composition—Fold1 and Fold3 are majority-negative, Fold0 and Fold4 are majority-positive, and Fold2 is nearly balanced (97/94)—and is not attributed to prevalence without further analysis. Continuous-risk R^2 (0.525 ± 0.112) is a positive blocked signal at this scale. This is reported as a robustness check on the protocol, not as citywide skill.

7 Discussion

7.1 Interpretation under the stated boundaries

The two pilots indicate that an open-label H3 table can be assembled, trained, and evaluated with blocked cross-validation; that mean rollups substantially alter fine-hotspot membership; and that trained-score adaptive refinement reduces uniform-fine cell count. These results demonstrate execution of the protocol under the stated pilot design; they do not establish product-ready city maps or rainfall-responsive screening under the current synthetic rainfall hook. On classification, the two pilots differ. The smaller table does not beat its constant-class baselines on accuracy or F1, whereas the expanded table beats the constant majority-class baseline on accuracy (0.642 versus 0.521) but not on F1 against the always-positive comparator (0.608 versus 0.641). Out-of-fold ROC-AUC and average precision are moderate in both pilots (pooled ROC-AUC 0.68 and 0.70; pooled PR-AUC 0.86 and 0.72), indicating some threshold-independent ranking discrimination that is stronger, relative to its prevalence baseline, in the more balanced expanded extent. Positive-class F1 nevertheless remains below the always-positive comparator in both pilots, so the discrimination evidence is treated as moderate rather than strong, and no citywide classification skill is claimed.

Relative to Svelling et al. (2026), the comparison is conceptual: they aggregate a proprietary building index into H3 for scalable communication, whereas this study learns on public labels with spatial holdouts and an explicit non-PFIb $PFI_h(c, r)$. The open-label Jaccard of 0.167 under mean aggregation (R10 to R8) is not narrated as matching their PFIb Jaccard of 0.14; the proximity is coincidental, given the different labels, resolutions, and hotspot definitions.

7.2 Limitations

1. **Spatial extent.** Results use two Manhattan pilots—Lower Manhattan ($n = 141$) and an expanded extent ($n = 956$)—neither of which is citywide New York City.
2. **Label bias.** 311 and related open indicators reflect reporting and mapping processes, not complete ground-truth inundation.
3. **Hydrographic proxy.** Distance-to-water from NHDPlus in a tidal and shoreline setting is a proxy, not inland drainage density.
4. **Rainfall provenance.** Event rainfall remains a synthetic constant hook; gauge or radar event ingestion is not yet implemented.
5. **Flat rainfall response.** The within-cell range of PFI_h across rainfall scenarios is currently 0; rainfall-conditioned discrimination is not demonstrated.
6. **Class imbalance and discrimination.** The smaller pilot is highly imbalanced (80% positive), and its accuracy and positive-class F1 fall below the always-positive majority baseline (0.808 / 0.893 versus 0.784 / 0.866). The expanded pilot is near-even and beats the constant majority-class baseline on accuracy (0.642 versus 0.521) but not on F1 against the always-positive comparator (0.608 versus 0.641). Out-of-fold ROC-AUC and PR-AUC are moderate in both pilots, so discrimination is supported at a modest level but is not claimed as strong skill.
7. **Small blocked design.** The smaller pilot distributes only seven H3 blocks over five folds (per-fold test 21–49, some folds a single block); the expanded pilot uses 28 blocks (per-fold test 190–193), which is more defensible but still a limited, non-citywide design.
8. **Continuous skill.** Spatial cross-validation R^2 is near zero in the smaller pilot (0.030) and moderate in the expanded pilot (0.525); the latter is reported as a scale-sensitive signal, not as citywide predictive skill.
9. **Held-out sensors.** FloodNet validation is unavailable under the default stub/opt-in path.

Random-split accuracy must not displace spatial cross-validation in claims. Fold-level variance (including Fold4 on $n = 24$ in the smaller pilot) further cautions against over-interpreting a single pilot run.

7.3 Outstanding steps

Future work should (i) ingest observed event rainfall with non-synthetic provenance; (ii) produce non-flat scenarios, that is, a within-cell PFI_h range greater than zero across rainfall intensities on the same static features; (iii) extend to a citywide or larger profile under the same spatial cross-validation protocol; and (iv) perform FloodNet held-out validation when a non-empty sensor layer is available.

8 Conclusions

This study presents a reproducible open-label H3 and machine-learning protocol—spatial block cross-validation, scale-loss diagnostics, adaptive cell-count refinement, and an explicit non-PFIb $PFI_h(c, r)$ definition—evaluated on two Manhattan open-data pilots. The live metrics demonstrate execution of the protocol under the stated boundaries but do not establish citywide operational skill. Out-of-fold ROC-AUC and average precision indicate moderate ranking discrimination, stronger relative to chance in the more

balanced expanded extent, while positive-class F1 remains below the always-positive comparator in both pilots. Observed event rainfall, a non-degenerate rainfall response, a full citywide extent, and FloodNet validation remain to be addressed.

Figure captions

Figure 1. Open-label H3 pluvial flood learning protocol (workflow). Conceptual pipeline. (1) Open multi-source inputs: flood labels (DEP stormwater polygons, 311 flooding points, USGS Ida high-water marks), static predictors (elevation, slope, impervious fraction, building density, distance-to-water), and a rainfall condition r (a synthetic constant hook in this pilot, not radar). (2) H3 assembly at R9 with provenance tags (assembly_mode, feature_source, label_source, rainfall_source). (3) Learning and blocked evaluation: a gradient-boosting classifier with a continuous-risk regressor, H3-block GroupKFold spatial cross-validation as the primary blocked evaluation, constant-class baselines (always-positive and always-negative), and logistic/ponding-rule baselines. (4) Diagnostics and outputs: the rainfall-conditioned $PFI_h(c, r)$ index (a definition/interface in this pilot, with currently flat scenario response), a scale-loss Jaccard ladder (R10 to R9/R8), adaptive refinement screened by trained PFI_h (to R11), and a Sandy negative-control check. The FEMA Sandy layer is a dashed side-channel that bypasses learning and enters only the negative-control diagnostic; it is never a training label.

Figure 2. Spatial H3-block cross-validation fold metrics. Per-fold classification accuracy and F1 on the Lower Manhattan pilot ($n = 141$; 5 folds; 7 blocks). Primary reporting uses the mean $\pm$ standard deviation across folds; individual high folds (e.g., Fold4) are not interpreted as citywide skill, and accuracy/F1 are reported alongside constant-class baselines.

Figure 3. Open-label scale loss across H3 resolutions. Jaccard similarity and F1 between fine-resolution (R10) hotspot parent sets and coarse (R8/R9) hotspot sets under mean, maximum, and p90 aggregation. Mean aggregation at R8 yields Jaccard of about 0.167, consistent with smoothing of local extremes; maximum and p90 preserve extrema by construction. These are diagnostics on the Lower Manhattan open-label pilot and must not be equated to Svelling et al.'s PFI_b Jaccard of 0.14.

Figure 4. Adaptive refinement versus uniform fine grids (cell counts). Fixed coarse (R9), adaptive mixed, and uniform fine (R11) cell counts for the pilot table (adaptive/uniform ≈ 0.569 ; 79 of 141 parents refined). The comparison is restricted to cell counts; it is not a wall-clock or citywide runtime claim.

Data and code availability

The public repository (code, configs, tests, paper documentation, and small summary tables; large rasters, geojson, trained model binaries, and large parquet files are excluded) is available at <https://github.com/Coucou2016/pluvial-flood-risk-DGGS-H3>. Layer provenance and download manifests are released with the repository. Synthetic demonstrations are excluded from scientific evidence. A process-

oriented research report and a data-authenticity audit document accompany the manuscript in the same documentation folder.

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Additional venue-specific references to be added.